A Feature Decomposition-Based Method for Automatic Ship Detection Crossing Different Satellite SAR Images

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***Abstract*— In the face of synthetic aperture radar (SAR) image object detection with different distributions of training and test data, traditional supervised learning methods cannot achieve good detection performance. Domain adaptation (DA) method has been shown to have the ability to solve this problem, but existing DA object detection algorithms all use adversarial DA theory for the detection task, which is ineffective in solving object regression localization in the detection task. In this article, to better solve the above problem, an automatic SAR image ship detection method based on feature decomposition crossing different satellites is proposed. The feature extraction layer of backbone network is divided into low level and high level, where domain-invariant feature (DIF) extractors are designed for the local features extracted from the low level and the global features extracted from the high level, respectively. We argue that the local and global features extracted from source domain and target domain contain domain-specific features (DSF) for adversarial DA and DIFs that contribute to object regression localization. Then, we decompose the local features and global features into DSF and DIF via vector decomposition method. For DSF counterpart, we introduce adversarial DA attention for feature alignment. DIF from the local features are fused into the backbone network for high-level global feature extraction. Finally, using region proposal network and adversarial domain classifier, we can get the accurate bounding box and object class of SAR image objects. Extensive experiments prove that the proposed method outperforms the state-of-the-art methods in terms of detection performance.**

***Index Terms*— Domain adaptation (DA), feature decomposition, object detection, regression localization, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) has a widespread application in urban planning and ocean monitoring because of its ability to capture rich scattering information by transmitting and receiving electromagnetic waves under all-weather conditions [1], [2], [3]. SAR image object detection has attracted much attention for its promising applications, such as ocean surveillance, road planning, and airport construction [4], [5], [6].

Traditional SAR image object detection methods include constant false alarm rate (CFAR) [7], extended fractal (EF) [8], etc. CFAR completes the detection of an object pixel by comparing the gray value of a single pixel with the detection threshold, which is determined by the statistical characteristics of the background clutter under a certain false alarm rate [7]. EF fully integrates the grayscale information of the image and the spatial distribution information of the grayscale, and uses the spatial difference of the reflected energy of the object and the clutter for object detection [8].

Since 2012, deep convolutional network has achieved a huge success on reaching high accuracy in classification task and has been quickly applied to object detection task [9]. Deep learning has taken the lead in demonstrating good problem-solving capabilities in fields such as natural images [10], [11] and optical remote sensing [12], [13]. Considering that SAR is a kind of remote sensing, deep learning technology has gradually begun to show its prominence in SAR image interpretation tasks [14], [15].

Despite the difference between SAR images and optical images, feature extraction from the raw data is always a critical prior step in object detection problem. Therefore, given the success achieved in the field of optical image object detection, deep learning is undoubtedly one of the most promising approaches for developing advanced SAR image object detection algorithms [14]. The current deep learning-based SAR image object detection methods mainly focus on the application of generalized detection networks under supervised learning to solve the detection problem, which mainly relies on the dataset of conclusive labels [15]. Simultaneously, many existing deep learning methods can only achieve good detection results when both the training and test sets come from the same SAR image dataset.

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With the increasing number of SAR imaging platforms in the world, there are more and more high-resolution SAR image datasets that researchers can use for further studies [16], [17]. As a result, in most cases, the problem we must address is SAR image object detection with different distributions between the training and test sets due to the differences in radar settings, imaging modes, viewing angles, scenes, and polarization patterns [18], [19], [20]. It is almost impossible for the existing methods to achieve the same detection performance as the supervised deep learning method by directly performing the detection task for the above situation. Besides, it is difficult to reacquire conclusive detection label for newly acquired datasets due to the specialized nature of SAR image label, poor visual friendliness, high labeling time cost and labor cost, and error-prone large-scale repetitive label. The development of domain adaptation (DA) [21], a branch of transfer learning, provides us with a solution to the above dilemma. The basic idea of DA is that since the source and target data distributions are different, the data are mapped into a feature space, where a metric is found to make the feature distributions of the source and target data as close as possible, so that the discriminator trained on the source domain features can be used on the target domain [22]. In DA, the labeled source domain data and the unlabeled target domain data are fed into the network for training, and then the trained model is tested on the target domain to conduct object detection task [23].

Existing methods for addressing the DA object detection problem are based on the adversarial DA theory and focus on aligning the feature-level distribution of the source and target domains to minimize the domain gap [18], [19], [20], [24], [25], [26], [27]. In the case of SAR images, the network finds it challenging to extract features from the grayscale images which can be used for detection task. Furthermore, when considering transfer between the source and target domains obtained from different imaging platforms, not all features are available for adversarial DA learning. Although many DA methods have been proposed in recent years, DA theory has been relatively less represented in detection task compared to classification task. This is due to the fact that the detection problem involves regression localization, while purely adversarial DA strategy is usually considered by discretizing the regression task into a number of interval classification tasks, and then applying the original DA theory to the classification problem. The interval discretization scatter already tells us that target domain generalization error bound is proportional to the square of the number of classes [28]. The number of classes in the common regression problem is greatly increased, and hence the generalization error constraint is significantly increased [29].

To make the features extracted by the network more suitable for DA object detection task, Liu *et al.* [30] divide the common and unique features in the source domain data and the target domain data into domain-invariant features (DIFs) and domain-specific features (DSF). The existing DA SAR image object detection methods do not consider DIF and DSF, but only the domain alignment of the features in both domains [18], [19], [20]. However, such an approach inevitably lacks the analysis of SAR image itself and the understanding of the regression task in DA object detection task. For example, in SAR image ship detection, the classes of ships that can be obtained by satellite-borne SAR are broadly classified as Cargo, Container, Fishing, Tanker, Others, etc. There may be some semantic differences between different classes of ships in these different datasets, but they may be similar in some features such as contours or textures, which means that when transferring between datasets with different distributions, we need to extract these features with differences for feature alignment to achieve accurate determination of who is the ship object in the target domain SAR images. In the regression problem, however, the distribution of features is discrete and diffuses throughout the space, then even if domain discriminator (or domain classifier) plays its role, there is still no guarantee that the features in the source and target domains are approximated according to the same labels. This makes that the prior feature alignment-based adversarial DA methods are not able to solve the regression localization problem in object detection well.

In general, with the increase of satellite-borne SAR imaging platforms, we can obtain more SAR image datasets, but it is very time-consuming and labor-intensive to relabel and train them. The DA method can be initially applied to the object detection problem of different satellite-borne SAR images, but because most of them copy the methods in natural images, they do not propose the DA network for the characteristics of the SAR image itself. Considering the SAR imaging system and SAR image features, it is necessary to propose a DA method that can meet the object class adversarial learning and object regression localization learning in large SAR image scenes.

To address the above problem, this article proposes an automatic SAR images ship detection method based on feature decomposition crossing different satellites. First, the backbone network feature extraction layer is divided into two parts: low-level feature extraction layer and high-level feature extraction layer. Furthermore, we argue that the local features extracted from the low level and the global features extracted from the high level are both consist of DIF and DSF, so DIF and DSF feature decomposition network modules are proposed to solve the challenges raised by DA detection task mentioned in the last paragraph. Specifically, in DIF and DSF feature decomposition modules, two DIF extractors (DIFEs) with different parameters are first designed to extract DIF from the local and global features. After decomposing DIF and DSF from local and global features, we construct the corresponding loss functions separately to further ensure that they are independent of each other. Then, since the local and global features only consist of DIF and DSF, DSF are obtained directly via feature vector decomposition method. For low-level DIF and DSF feature decomposition, DSF are then used in the adversarial DA attention subnetwork for instance-level feature domain alignment, while DIF of local feature return to the backbone network for high-level feature extraction. For high-level DIF and DSF feature decomposition, DIF are used for regression localization of SAR image object because they contain features that facilitate the extraction of region of interest (RoI). The obtained DSF are fed into the adversarial DA attention subnetwork to obtain instance-level features,

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which are fused with the instance-level features obtained at the low level. The fused instance-level features are then subjected to the operation of adversarial domain feature alignment via gradient reversal layer (GRL) [21]. Here we do not perform the same pixel-level feature alignment operation as most of the existing DA object detection methods, because we have already extracted DSF for feature alignment and do not need as many complicated steps. Finally, the results of RoI alignment by region proposal network (RPN) and domain classifier by adversarial DA are obtained to achieve the object detection task crossing two SAR image ship datasets with different distributions.

In summary, the main contributions of this article can be summarized as follows.

1) Due to the limitations of the adversarial DA method in the SAR image object detection task, we propose to consider DIF and DSF in different distributions of SAR image object feature space, where DSF are the ones that are meaningful for feature alignment in the adversarial DA method. As a result, an overall SAR image DA object detection network framework based on feature decomposition is constructed.
2) Feature decomposition is introduced after both low-level features and high-level features extraction in the backbone network, and the local features at the low level and global features at the high level are divided into DIF and DSF by means of vector decomposition, respectively. The vector decomposition method has less computationally intensive and makes DIF and DSF more independent than the existing extraction of DIF and DSF by two feature extractors. In addition, two loss functions are proposed to further make DIF and DSF independent, respectively.
3) Extensive experiments are conducted on SAR image ship datasets obtained from different satellite-borne SARs, and the experimental results clearly show that the proposed method outperforms the existing methods' state-of-the-art (SOTA) DA method by more than 2% in terms of SAR image object detection accuracy for different satellites. Moreover, the DIF and DSF decomposition idea in our proposed method do facilitate DA SAR image object detection via multiple ablation studies.

The remainder four sections of this article are organized as follows. Section II recaps the related work of SAR image object detection and DA object detection. Section III introduces the proposed DA framework for SAR image object detection. Experimental results and corresponding analysis based on two SAR image datasets from different satellite-borne SAR imaging platforms are given in Section IV. Finally, Section V concludes this article.

II. Related Work

A. SAR Image Object Detection

1) Traditional SAR Image Object Detection: Traditional SAR image object detection aims at obtaining the object position via utilizing detection systems combined independent functional components in a sequential cascade [31]. A method for extracting ship features from SAR images is proposed, using a standard algorithm to generate a closed contour image of the ship, a two-step method to accurately estimate the orientation angle, and a modified centerline and minimum outer rectangle extraction criterion [32]. To address the problem that the traditional CFAR detection algorithm is difficult to distinguish and detect the neighboring SAR image objects, Hou *et al.* [33] use a multilayer CFAR SAR image ship detection algorithm with lognormal distribution to reduce the influence of weak-scattering points on the detection results and improve the detection capability of CFAR method through the multilayer CFAR method, which is more effective in ship object detection. Abdelkawy and Mcgaughy [34] propose the use of exponential wavelet fractal feature extraction for object detection, which has the dual effect of enhancing edge pixels and smoothing uniform regions due to the peak sharpness and smoothness characteristics of the exponential factor.

2) Deep Learning SAR Image Object Detection: SAR image object detection based on deep learning has become a popular topic in recent years, thanks to the potential to achieve better accuracy and manage bigger amounts of data through the extraction of object features [14]. Prior studies using deep networks to detect ships involved two steps: clipping image slices from the overall SAR image and then employing a convolution neural network (CNN) to assess whether the image slice contained an object [35], [36]. The problem of poor image slice size adaptation can be effectively solved using faster region CNN (RCNN) [37] with subsequent short-joining feature fusion techniques [16], [38], [39]. Li *et al.* [16] fuse the features of the last three convolutional layers together and then feed them into a candidate region generation network to improve the ship detection performance. To make full use of different resolution features, Jiao *et al.* [40] tightly connect each layer of the backbone network to its subsequent layers and pass all the convolutional layer features into each other independent of RPN to generate candidate edge, thus improving the candidate edge quality. Rostami *et al.* [41] and Huang *et al.* [42] explore knowledge transfer- and domain transfer-based SAR data domain transfer learning studies for SAR image object detection.

B. DA Object Detection

When the training and test sets are datasets obtained from different satellite-borne SAR imaging platforms, DA is an excellent solution for data with different distributions between labeled source domain and unlabeled target domain. To improve the cross-domain robustness of object detection, Chen *et al.* [43] address domain shift at two levels: image-level shift, such as image style, lighting, etc., and instance-level shift, such as object appearance, size, etc. Saito *et al.* [44] propose a method for detector adaptation based on strong local alignment and weak global alignment, which focuses the adversarial alignment loss on globally similar images with less emphasis on images that are globally different for the weak alignment model, and looks at only the local perceptual field of the feature map for the strong domain alignment model. Rezaeianaran *et al.* [25] propose an

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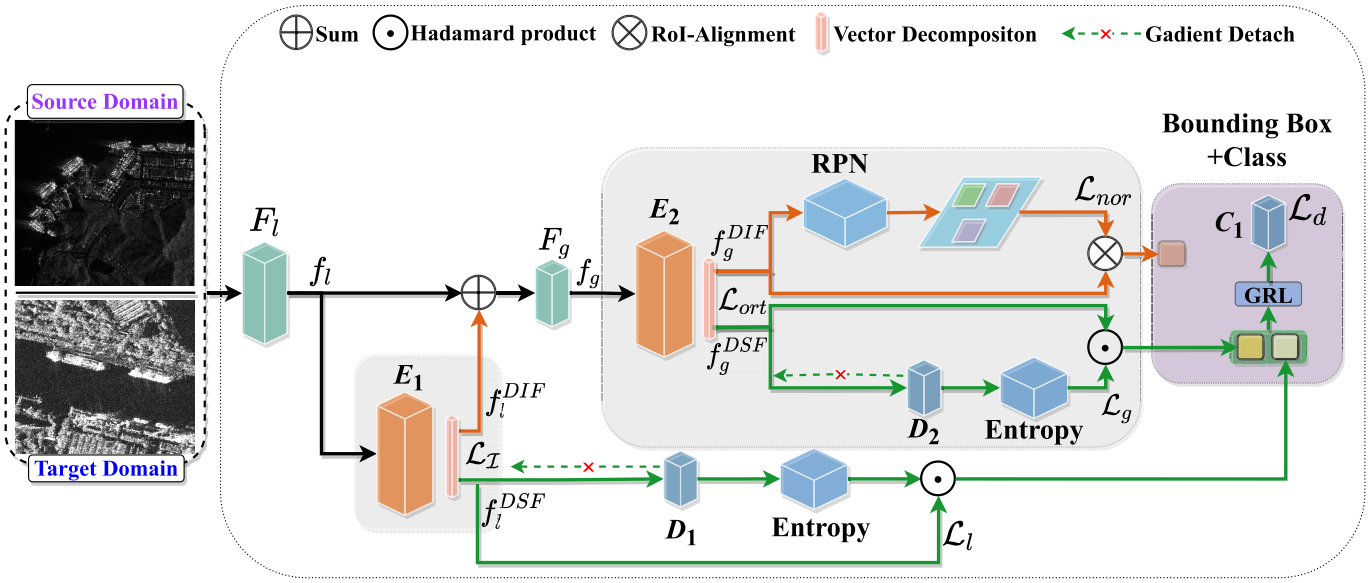


Fig. 1. Proposed DA SAR image object detection framework. The labeled source domain and unlabeled target domain are two SAR datasets with different distribution. F_l and F_g are two feature extractors of the backbone network. E_1 and E_2 are two DIFEs. D_1 and D_2 are two domain discriminators for adversarial learning. Entropy is the operation for calculating entropy vectors of feature vectors. "GRL" is gradient reversal layer and C_1 is a domain classifier. Some other relevant shapes and their meanings are shown in the upper blank part.

unsupervised DA (UDA) algorithm that aggregates features at the instance level based on visual similarity before inducing group alignment by adversarial training. VS *et al.* [45] use class discriminators to ensure class-aware feature alignment to learn domain-invariant discriminative features and suggest the generation of memory-guided class-specific attention graphs, which are then used to route features appropriately to the corresponding class discriminators.

III. Proposed DA SAR Image Object Detection Method

For the situation of SAR image object detection obtained from different satellite-borne SAR imaging platforms, the labeled source domain $\mathcal{S} = \{x_i^{\mathcal{S}}, b_i^{\mathcal{S}}, y_i^{\mathcal{S}}\}_{i=1}^{N_{\mathcal{S}}}$ and the unlabeled target domain $\mathcal{T} = \{x_i^{\mathcal{T}}\}_{i=1}^{N_{\mathcal{T}}}$ are defined. In $\mathcal{S}$ and $\mathcal{T}$, we define x_i is the ith image in SAR dataset which has N images in total. Besides, since source domain is labeled, $b_i^{\mathcal{S}}$ and $y_i^{\mathcal{S}}$ denote the bounding box and class label of $x_i^{\mathcal{S}}$, respectively.

Faster RCNN is utilized as the baseline network whose training input is a combination of labeled source domain SAR images and unlabeled target domain SAR images. The overall objective is to achieve good detection results on target domain SAR images. The feature extraction portion of the baseline network is divided into two levels: local feature extraction on the low level and global feature extraction on the high level. Specifically, Fig. 1 presents the overall DA object detection framework.

A. Decomposition Network Module of DIF and DSF

Totally, we first propose a DA object detection framework based on Faster RCNN baseline network. The feature decomposition network modules of DIF and DSF are introduced after the low-level feature extraction and the high-level feature extraction of the backbone network, respectively. Specifically,

TABLE I
Network Structure of E_1

Layer	Function	Hyperparameters
1	Convolution	in=1024, out=512, k=3, s=1, p=1
2	BatchNormal, ReLU	num=512
3	Dropout	0.5
4	Convolution	in=512, out=512, k=3, s=1, p=1
5	BatchNormal, ReLU	num=512
6	Dropout	0.5
7	Convolution	in=512, out=1024, k=3, s=1, p=1
8	BatchNormal	num=512

as an input SAR image from source domain or target domain, x_i is fed into the first feature extractor F_l. The local feature map vector $\{f_l^{H\times W\times C}\}_i = F_l(x_i)$ can be thereby obtained, where H, W, and C are the height, width, and channel numbers of the feature map, respectively. In object detection task, we generally do not consider the channel numbers, so for simplicity we will omit C and write the local feature map vector directly as $f_l^{H\times W}$.

For the local features $f_l^{H\times W}$ of SAR image ship object in source domain and target domain, we argue that they contain DIF such as ship contour and texture as well as DSF such as deep semantic features. To extract the DIF from the local features, we design a DIFE E_1 inspired by [30], whose network structure is shown in Table I, where in, out, k, s, p, and num denote the input channel numbers, output channel numbers, kernel size, stride, padding of convolution layer, and feature channel number of BatchNormal layer, respectively. Therefore, DIF of local feature map can be obtained as follows:

$$f_l^{\mathrm{DIF}} = E_1(f_l). \tag{1}$$

In convolution neural network (CNN), the features extracted by the network are in the form of feature vectors. Higgins *et al.* [46] have proven that for vector disentanglement representation, if the two vectors obtained by vector

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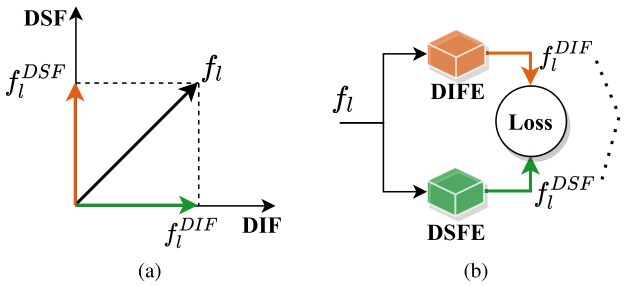


Fig. 2. Decomposition method of DIF and DSF. (a) Vector decomposition, where blue, yellow, and green arrows denote f_l^{DIF}, f_l^{DSF}, and f_l, respectively. (b) Previous decomposition, where DIFE and DSFE are separately used to extract DIF and DSF, and the dashed line indicates the process of making DIF and DSF independent through a loss function.

decomposition are orthogonal, they are least affected from each other. Since we consider that the feature vectors extracted by the backbone network only consist of DIF and DSF, vector decomposition strategy is used to separate DIF and DSF. Unlike previous methods that require another DSF extractor (DSFE) to obtain the DSF and then gradually make the DIF and DSF independent of each other via a loss function [30], [47], we directly consider DIF and DSF as two mutually orthogonal feature vectors and obtain DSF by vector decomposition

$$f_l^{\mathrm{DSF}} = f_l - f_l^{\mathrm{DIF}} \tag{2}$$

where $f_l^{\mathrm{DIF}} \perp f_l^{\mathrm{DSF}}$. It is important to note that since DIF are extracted from the local feature by DIFE and are computed with numerical value and directional dimensions, the vector decomposition can make DIF and DSF orthogonal. Furthermore, this process of decomposing the vector into two components that are mutual orthogonal can be seen in Fig. 2(a). Vector decomposition, different from traditional disentanglement in Fig. 2(b), simply takes the difference between the original input and the decomposed component as the last component, lowering parameters and computing expense. In addition, constructing DSFE does not guarantee that the extracted DSF and DIF do not have overlapping feature vectors, which is not useful to the object detection task as we analyzed in Section I.

To make the domain-invariant and domain-specific portions of the local features contain more information about their counterparts, we consider introducing mutual information [48] of two variables to further qualify domain-invariant information and domain-specific information mutual independence

$$\mathcal{I}(M;N) = \int_{M\times N} \log \frac{d\mathbb{P}_{MN}(m,n)}{d\mathbb{P}_M \otimes d\mathbb{P}_N} d\mathbb{P}_{MN}(m,n) \tag{3}$$

where $\mathbb{P}_{MN}(m,n)$ is the joint probability distribution of two variables, $\mathbb{P}_M$ and $\mathbb{P}_N$ indicate the marginal probability of two variables, respectively. Here we use the average pooling results of DIF and DSF as domain-invariant information and domain-specific information. The mutual information captures the nonlinear statistical dependencies between variables and can thereby be used as a measure of true dependence. Following the interpretation of the mutual information and Eq. (3), we employ the Kullback-Leibler (KL) divergence of the joint probability distribution and the product between two marginal distributions as the mutual information between DIF and DSF

$$\mathcal{L}_{\mathcal{I}}(M;N) = D_{KL}(\mathbb{P}_{MN} \| \mathbb{P}_M \otimes \mathbb{P}_N). \tag{4}$$

It is intuitive from (4) that the larger the divergence between the joint probability distribution and marginal distributions product, the stronger the dependence between DIF and DSF. When the two features become fully independent of each other, this divergence, also known as mutual information, vanishes.

We have extracted DIF of the local features on the low-level network in the above step. However, this does not guarantee that DIF of both the local features and the global features can be used to implement a DA detection task. Therefore, we further study the DIF and DSF decomposition in global features extraction. Through a summation process, the feature used as input to the high-level feature extractor F_g can be written as follows:

$$f_1 = f_l + f_l^{\mathrm{DIF}}. \tag{5}$$

Then, the global feature is obtained by

$$f_g = F_g(f_1). \tag{6}$$

For the extracted global feature vector, we also propose a DIFE E_2 to extract the DIF in it and obtain the DSF by vector decomposition

$$f_g^{\mathrm{DIF}} = E_2(f_g) \tag{7}$$

$$f_g^{\mathrm{DSF}} = f_g - f_g^{\mathrm{DIF}} \tag{8}$$

where $f_g^{\mathrm{DIF}} \perp f_g^{\mathrm{DSF}}$. The network structure of E_2 is similar to that of E_1, which is shown in Table II, where in, out, k, s, p, and *num* denote the input channel numbers, output channel numbers, kernel size, stride, padding of convolution layer, and feature channel number of BatchNormal layer, respectively.

TABLE II
NETWORK STRUCTURE OF E_2

Layer	Function	Hyperparameters
1	Convolution	in=256, out=128, k=3, s=1, p=1
2	BatchNormal, ReLU	num=128
3	Dropout	0.5
4	Convolution	in=128, out=128, k=3, s=1, p=1
5	BatchNormal, ReLU	num=128
6	Dropout	0.5
7	Convolution	in=128, out=256, k=3, s=1, p=1
8	BatchNormal	num=256

Since the extracted DIF are useful for the bounding box regression of SAR image object, RoI-Alignment proposals of f_g and f_g^{DIF} should be as similar as possible for the same object at the same location. Specifically, pooling results $p_g^{\mathrm{DIF}} \in \mathbb{R}^{P\times C\times H\times W}$ and $p_g \in \mathbb{R}^{P\times C\times H\times W}$ can be obtained via RoI-Alignment results of f_g^{DIF} and f_g, where P, C, H, and W denote proposal numbers, channel numbers, height, and width, respectively. Furthermore, row adjacency matrices are defined via softmax process

$$\begin{aligned} \mathbf{A}_g^{\mathrm{DIF}} &= \mathrm{softmax}_{\mathrm{row}}\left[\left(p_g^{\mathrm{DIF}}\right)\left(p_g^{\mathrm{DIF}}\right)^{\mathrm{T}}\right] \\ \mathbf{A}_g &= \mathrm{softmax}_{\mathrm{row}}\left[\left(p_g\right)\left(p_g\right)^{\mathrm{T}}\right] \end{aligned} \tag{9}$$

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where $\text{softmax}_{\text{row}}(\bullet)$ is softmax process across the row direction. The objective function is defined as the L_2-norm of these two adjacency matrices

$$\mathcal{L}_{\text{nor}} = \left\|\mathbf{A}_g - \mathbf{A}_g^{\text{DIF}}\right\|_2^2. \tag{10}$$

For the extracted DIF, RPN is performed to extract the object proposals. We have previously decomposed the global feature f_g into f_g^{DIF} and f_g^{DSF} via vector decomposition, and then an orthogonal objective function is defined to qualify the orthogonality of these two features. Concretely, though f_g^{DSF} does not be set as input to RPN, the pooling result $p_g^{\text{DSF}} \in \mathbb{R}^{P\times C\times H\times W}$ results can be also obtained. Therefore, the orthogonal objective function can be written as follows:

$$\mathcal{L}_{\text{ort}} = \frac{1}{n}\sum_{x=1}^{n}\left|\sum_{y=1}^{c}\mathbf{M}(x,y)\right| \tag{11}$$

where $\mathbf{M} = \left\|p_g^{\text{DIF}}\right\|_2^2 \odot \left\|p_g^{\text{DSF}}\right\|_2^2$ indicates the value of $\mathbf{M}$ at position (x, y), $|\bullet|$ and $\odot$ denote the absolute value and Hadamard product, respectively. The DIF and DSF become more independent of each other as the objective function $\mathcal{L}_{\text{ort}}$ is gradually minimized. Simultaneously, the objective function $\mathcal{L}_{\text{ort}}$ can make DSF and DIF contain more domain-specific information and domain-invariant information, respectively. In addition, it is worth noting that we use the alignment results rather than the overall feature map to calculate the orthogonal loss, which not only reduces the computational cost but also promotes our model to focus on SAR object location.

B. DA Feature Decomposition Object Detection

Since DIF and DSF have been already decomposed in the previous steps, according to our analysis, DIF are further used for regression localization in the detection task, while DSF are used for class feature alignment. The regression localization task for DA object detection mainly focuses on obtaining the specific location of the object bounding box on the target domain using the foreground proposals of the labeled source domain $\left\{x_i^{\mathcal{S}}, b_i^{\mathcal{S}}\right\}_{i=1}^{N_{\mathcal{S}}}$ and the unlabeled target domain $\left\{x_i^{\mathcal{T}}\right\}_{i=1}^{N_{\mathcal{T}}}$. The goal of the class feature alignment task for DA object detection, on the other hand, is to obtain accurate target domain proposals using the labeled source domain $\left\{x_i^{\mathcal{S}}, y_i^{\mathcal{S}}\right\}_{i=1}^{N_{\mathcal{S}}}$ and the unlabeled target domain $\left\{x_i^{\mathcal{T}}\right\}_{i=1}^{N_{\mathcal{T}}}$.

After feature vector decomposition we have proposed, the dimensions of both f_g^{DIF} and f_g^{DSF} are set to be the same as the dimension of f_g. RPN is executed on f_g^{DIF} to extract a set of domain-invariant proposals, which are used for the localization task of SAR image objects. Thus, the detection objective function in the network structure can be defined as

$$\mathcal{L}_{\text{det}} = \begin{cases} \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}}, & \text{Source Domain} \\ \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cls}}, & \text{Target Domain} \end{cases} \tag{12}$$

where $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{cls}}$ separately indicate the bounding box regression loss function and the object classification loss function. Since the target domain is unlabeled, $\mathcal{L}_{\text{rpn}}$ denotes the loss function to distinguish between foreground and background, and refines the bounding box which is only employed in source domain.

For DSF, we use the proposed adversarial learning attention mechanism [18] to achieve more direct adversarial DA learning when performing feature alignment operation. On the multilevel feature representation such as local and global features of the backbone network, we expect that the domain-specific information about the object of interest can be enriched by contextual instance-level feature alignment. Specifically, we feed f_l^{DSF} and f_g^{DSF} on both low level and high level of the backbone network into the instance-level contextual feature domain adversarial learning attention. Unlike existing studies on DA object detection about both DIF and DSF [30], [47], we employ gradient detach instead of GRL here. Gradient detach prevents gradients from flowing from the context subnetwork through the detection backbone path, helping to obtain more distinguishing contexts and can lead to more diverse information; at the same time, models that have used this strategy pay less attention to the context compared to ordinary models. Thus, this can prevent the domain-invariant information from being influenced by the gradient of the DSF domain adversarial learning process, resulting in more accurate class feature alignment.

In the low-level local feature extraction stage, we propose a domain discriminator to implement domain alignment operation

$$d_l = D_1\left(\text{GD}\left(f_l^{\text{DSF}}\right)\right) \tag{13}$$

where $\text{GD}(\bullet)$ denotes gradient detach operation.

Then, the entropy vector is calculated from the probability output of (13) as follows:

$$e_l = -d_l \odot \log(d_l). \tag{14}$$

Next, the calculated entropy vector is fused into the feature vector, which allows the high- and low-entropy regions in the SAR image to be well distinguished

$$w_l = f_l^{DSF} \odot (1 + e_l). \tag{15}$$

Here, an objective function is proposed to constrain the above domain adversarial learning attention as follows:

$$\mathcal{L}_l = \frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_j \log\left(\left(\{e_l\}_i^{\mathcal{S}}\right)_j\right)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_j \log\left(1 - \left(\{e_l\}_i^{\mathcal{T}}\right)_j\right)^2. \tag{16}$$

For the f_g^{DIF} extracted on the high level, we introduce a similar adversarial learning attention mechanism. We perform domain adversarial learning by the same steps as (11)–(13)

$$d_g = D_2\left(\text{GD}\left(f_g^{\text{DSF}}\right)\right) \tag{17}$$

$$e_g = -d_g \odot \log\left(d_g\right) \tag{18}$$

$$w_g = f_g^{\text{DSF}} \odot \left(1 + e_g\right). \tag{19}$$

Similarly, the objective function can be written as follows:

$$\mathcal{L}_g = \frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_j \log\left(\left(\left\{e_g\right\}_i^{\mathcal{S}}\right)_j\right)^2 + \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_j \log\left(1 - \left(\left\{e_g\right\}_i^{\mathcal{T}}\right)_j\right)^2. \tag{20}$$

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Furthermore, we integrate the instance-level features obtained at two levels into a new feature vector, represented as

$$w_n = [w_l, w_g] \tag{21}$$

where $[w_l, w_g]$ indicates the integration of the two feature vectors in series. Then, the above integrated feature vector is fed into the SAR object domain classifier C_1 after GRL which is to achieve domain adversarial learning. Finally, the objective function can be obtained as follows:

$$\mathcal{L}_d = -\frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_{i,j} c_i \log\left(\left(\{c_n\}_i^{\mathcal{S}}\right)_j\right) - \frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_{i,j}(1-c_i)\log\left(1-\left(\{c_n\}_i^{\mathcal{T}}\right)_j\right) \tag{22}$$

where c_i is the domain label of ith training image with $c_i = 1$ for the source image and $c_i = 0$ for the target image, and $c_n = C_1(w_n)$.

C. Overall Training Objective

This section demonstrates the proposed overall training objective function for DA SAR image object detection based on feature decomposition. Faster RCNN is employed as the DA detection backbone network with DIF and DSF decomposition on the low level and high level of the network, respectively. The domain-invariant information is used for regression localization of object box while the domain-specific information is used for feature classification by adversarial domain learning attention mechanism.

In the DA object detection problem, feature extractors and domain discriminators are more likely to play a competitive game. For the extracted DSF, the task of the domain discriminator is to be responsible for identifying whether the features output by the feature extractor are from the source domain data or the target domain data. And the feature extractor has to minimize the difference between the source domain data and the target domain data output vector to "obfuscate" the domain discriminator from identifying it correctly.

Therefore, to train the whole DA object detection model, we construct the above-mentioned loss function as a complete objective function

$$\max_{D_1,D_2,C_1}\min_{F_l,F_g}\mathcal{L}_{\text{det}} + \lambda\left(\mathcal{L}_l + \mathcal{L}_g + \mathcal{L}_d\right) + \gamma\left(\mathcal{L}_{\text{nor}} + \mathcal{L}_{\text{ort}} + \mathcal{L}_{\mathcal{I}}\right) \tag{23}$$

where λ and γ are trade-off hyperparameters for balancing detection loss function, domain adversarial learning loss function, and feature decomposition loss function.

IV. Experimental Results and Analysis

In this section, we present in detail the experimental datasets, the experimental initialization settings, and the related experimental results. Two SAR image datasets obtained from different imaging platforms are mainly used in the experiments, which are Gaofen-3 dataset and SAR ship detection dataset (SSDD) [16]. Furthermore, a comparison experiment with the existing SOTA DA object detection method and a series of ablation studies will be implemented to demonstrate the effectiveness and superiority of the proposed method.

A. Datasets Description and Experiment Initialization

In this article, we select two SAR image object detection datasets with domain-shift, which are Gaofen-3 and SSDD. Gaofen-3 dataset is a satellite-borne SAR imaging result developed independently by China, which can be downloaded by specific researchers through open channels. Our group has collected and produced an SAR image object detection dataset for Gaofen-3 in recent years, including 1000 SAR images with an aspect ratio of 1:1. These images are cut from large-scale scene images with a resolution between 1 and 3 m to perform SAR image object detection task. Gaofen-3 ship detection dataset includes two major scenes: inshore ships and offshore ships where harbors, islands, and sea surfaces with varying conditions at sea level are among the scene types. Thousands of ships in 11 classes, such as Cargo ships, Transport ships, Tankers, and Fishing vessels, are among the objects. We refined the label of this dataset through professional manual interpretation and corresponding scene analysis on Google Earth, and the annotation format is the same as Pascal visual object classes (VOC) [49].

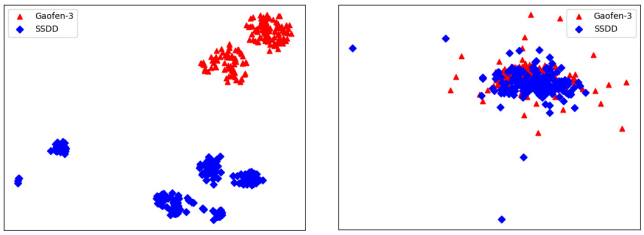
SSDD is first published by Li *et al.* [16] in 2017 at the conference of SAR in Big Data Era. SSDD is also the first available SAR image object detection dataset, a year and a half ahead of the second open dataset. Therefore, SSDD is also one of the most frequently used SAR image object detection datasets which is obtained by TerraSAR-X, RadarSat-2, and Sentinel-1 satellite-borne SAR imaging platforms. The SAR image samples in SSDD are diverse, with different resolutions from 1 to 15 m, different polarizations, different sea states, different ship scenarios including inshore and offshore, and different ship sizes. There are typical hard detection samples in SSDD. Each of these samples requires special consideration in practical applications of SAR ship detection, such as (1) small ships with imperceptible features, (2) dense parallel ships berthing at ports with overlapping hulls, (3) different classes of ships, (4) ship detection under severe scattering noise and multiple types of sea clutter, and (5) ship detection in complex situations. The specific technical details of the two datasets can be seen in Table III.

Concretely, these two datasets are obtained from different satellite-borne SAR imaging platforms. The former is imaged by Chinese C-band Gaofen-3, and the latter is the imaging result of German X-band TerraSAR-X, Canadian C-band Radarsat-2, and European Space Agency C-band Sentinel-1. The four satellite-borne SARs have different resolutions, incidence angles, polarization modes, imaging modes, and other characteristics. According to previous studies [18], [50], all of these characteristics result in different distributions in the two SAR imagery datasets. The different distributions of two sets of data are caused by various factors. In optical remote sensing images, the differences in illumination intensity, shooting angle and shooting time will cause different distributions of the images; while in SAR images, the differences in imaging

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TABLE III
COMPARISON OF TECHNICAL PARAMETERS BETWEEN GAOFEN-3 DATASET AND SSDD

Dataset	Gaofen-3	SSDD
Sensors	Gaofen-3	TerraSAR-X, RadarSat-2, Sentinel-1
Polarization	HH, VV, VH, HV	HH, VV, VH, HV
Resolution	1-3 m	1-15 m
Places	Tianjin, China; Changi, Singapore; Hongkong, China	Yantai, China; Visakhapatnam, India
Aspect ratio	1:1	1:1, 1:2, 2:1
Ship	11 classes in different sizes and materials	8 classes in different sizes and materials
Sea condition	Good and bad conditions	Good and bad conditions
Scenes	Inshore and offshore	Inshore and offshore
Image number	1000	1160
Ship number	3694	2456



(a) (b)

Fig. 3. Feature distributions of Gaofen-3 dataset and SSDD presented by t-SNE. The red triangle indicates Gaofen-3 dataset and the blue diamond indicates SSDD. The two dimensions in the figure are obtained by t-SNE downscaling which do not explicitly indicate the physical meaning, so they are omitted. (a) Feature distributions before DA. (b) Feature distributions after DA.

platforms, polarization methods, and imaging methods will make the two sets of data have different distributions. In DA object detection problem, we mention the different distributions more at the feature distribution level, and the original feature distributions of Gaofen-3 and SSDD datasets can be shown in Fig. 3(a) by t-distribution stochastic neighbor embedding (t-SNE) [51].

According to the above description, for DA SAR image object detection task, Gaofen-3 and SSDD datasets are two SAR image data domains with different distributions. Without loss of generality, we separately treat Gaofen-3 dataset and SSDD as the source domain and target domain in the course of subsequent experiments, and thereby define two DA SAR image object detection tasks: Gaofen-3 to SSDD (G $\rightarrow$ S) and SSDD to Gaofen-3 (S $\rightarrow$ G).

We randomly divided the datasets into a training set and a test set with a ratio of 7:3 in object detection experiment. Although the size of the dataset we used is not sufficient, unlike many existing studies, we do not perform other data expansion operations. Using the images from both datasets as input to the network, the network initialization reshape them to the appropriate size. Furthermore, based on both SAR imagery datasets, we separately calculate the mean and standard deviation of the data, then normalize all images by subtracting the mean from the uniform channel dimension to balance the gray value range for stable training. In both training and test, the batch size of the network is set to 1.

For the proposed DA SAR image object detection framework, we employ Faster RCNN as the baseline network and ResNet-101 as the pretrained backbone network. Similarly, the proposed network framework also supports visual geometry group (VGG)-16 as the backbone network, which will be demonstrated in the ablation experiment subsection later. Finally, we use Nonmaximum Suppression [52] to calculating the final detection results whose threshold is empirically set to 0.3. During the training process, the initial learning rate is set to 0.001 and decays by 0.1 every five epochs after the fifth epoch. For the training procedure, we use stochastic gradient descent (SGD) with a momentum of 0.9 as the optimization approach. For all experiments, the Intersection over Union (IoU) threshold is set to 0.5 to evaluate detection performance. The empirical IoU threshold of 0.5 is used by most existing SAR object detection algorithms, and we show the detection outcomes under various IoU thresholds in Section IV-C3. For the two hyperparameters in the objective function (23), we empirically set $\lambda = 1$ and $\gamma = 0.5$.

Several parameter indicators are used in the object detection task to describe whether the detection is correct or not. These parameter indicators are true positives (TP), false positives (FP), true negatives (FN), and false negatives (FN), whose meanings are predicted by the model as correct positive samples, FP samples, correct negative samples, and FN samples, respectively. To evaluate detection method quantitatively, we employ precision (PR), recall (RE), and mean average PR (mAP), which are represented as follows:

$$\mathrm{PR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{24}$$

$$\mathrm{RE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{25}$$

$$\mathrm{mAP} = \frac{1}{m} \sum_{m} \left[\int \mathrm{PR}(\mathrm{RE}) d(\mathrm{RE}) \right]. \tag{26}$$

PR indicates the proportion of correct predictions among all samples predicted by the model, while RE indicates the proportion of all true samples that the model predicts correctly. mAP is the area under the PR-RE curve, which also depicts the overall performance of the proposed detection method.

All of the experiments are performed on a GIGABYTE X299WU8 workstation with an Intel Core(R) i9-10920X central processing unit (CPU), an Nvidia GeForce RTX 3090 Turbo graphics processing unit (GPU), and 64 G RAM under Ubuntu 18.04. Our code is constructed on the Faster RCNN baseline network based on the Pytorch framework.

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TABLE IV
OVERALL SAR IMAGE OBJECT DETECTION RESULTS (%) OF VARIOUS METHODS

Method	G → S			S → G		
	PR	RE	mAP	PR	RE	mAP
Faster RCNN [37]	62.5	77.8	67.0	57.7	71.0	57.9
FACL [18]	74.6	82.9	78.1	69.8	79.9	68.4
UDA [19]	73.5	81.9	76.2	66.9	78.0	67.3
SDA [44]	68.6	77.2	70.1	63.7	75.3	61.9
SCL [26]	69.4	78.6	71.4	65.9	76.8	63.2
MeGA [45]	71.3	80.2	73.1	66.6	77.2	65.6
DTA [30]	75.8	83.1	79.2	70.4	80.3	69.2
I^3NET [47]	73.8	82.2	77.6	67.2	78.4	67.6
Ours	**78.4**	**86.3**	**81.5**	**73.7**	**81.9**	**74.4**

B. Comparison Experiments

In this section, the proposed DA SAR image object detection method is compared with conventional Faster RCNN [37] and other seven SOTA DA object detection methods. For Faster RCNN baseline network [37], to have the same initial setting as DA object detection problem, we use the labeled source domain dataset as the training set, and the trained model is tested on the target domain dataset. For the seven SOTA methods, five of them are feature alignment methods in SAR community and Computer Vision community based on adversarial DA [18], [19], [26], [44], [45], and the remaining two are methods that perform DIF extraction [30], [47]. In the following writing, we replace them with the abbreviated forms of these methods such as feature alignment and compensation loss (FACL) [18], UDA [19], stacked complementary losses (SCL) [26], strong-weak distribution alignment (SDA) [44], MeGA [45], decompose to adapt (DTA) [30] and I^3NET [47].

For two SAR image object detection tasks, G → S and S → G, the experiments are conducted individually, and the quantitative results of the comparison experiments are shown in Table IV. As can be seen from Table IV, the proposed method has the best detection accuracy for both detection tasks. For faster RCNN baseline network method, the model trained on the source domain cannot achieve good test accuracy on the target domain test dataset because there is no target domain dataset involved in the training process. Therefore, the DA method is helpful to improve the target domain detection accuracy. Meanwhile, Table IV also shows that the SAR image object detection accuracy of the feature alignment-based adversarial DA methods is relatively lower than that of the methods that extracts DIF, except for our previous work on FACL [18]. The specific reason is that I^3NET is customized based on one-stage detector using the natural features of different layers of deep features to learn implicitly the instance DIF instead of considering the DIF more conducive to the regression localization of the objects we have mentioned in this article. Specifically, the proposed method has a detection mAP at least 3.4% higher than the feature alignment-based adversarial DA method and at least 2.3% higher than the method based on extracting DIF. Table IV also shows that the SAR image object detection mAP for G → S task is generally higher than that for S → G task. Because comparing these two datasets, although SSDD contains more types of satellite-borne SAR images, the DA network still focuses on the features of the SAR images and the ship objects therein during the learning process. Gaofen-3 dataset, on the other hand, has more complex scenes, less background clutter, and more classes of ship objects, all of which improve the overall detection capability in all aspects when performing the detection task. Moreover, the proposed DA method is trained with the labeled source domain and the unlabeled target domain, which means that in the G → S task, the network is performing supervised learning on a relatively difficult dataset and unsupervised learning on a relatively easy dataset. Thus, the network achieves better DA capacity when Gaofen-3 dataset is used as the source domain, which results in better detection performance for G → S task than for the S → G task. Simultaneously, Fig. 3(b) shows the feature distribution results of the source and target domain datasets after DA, which indicates that the proposed method also performs well in feature alignment. In addition, we have selected some typical scenes of SAR image object detection results in the two tasks which are shown in Figs. 4-6, respectively.

In G → S task, we select two typical scenes for SAR image object detection to demonstrate the experimental results of the proposed method and the comparison methods, respectively. For the overlapping object detection on the sea surface in Fig. 4, due to the influence of sea clutter and noise, a distinct cross-sidelobe appears around the single ship object, which makes all feature alignment-based adversarial DA methods detect it as two objects. However, the methods of extracting DIF can well circumvent the problem of wrong detection caused by cross-sidelobe. Of course, it is also clear from Fig. 4 that none of the methods except ours accurately detects the rectangular box size of this object. For Fig. 4(a)–(f), the network has the problem of wrongly detecting the cross-sidelobe as object as well as failing to accurately detect the actual object bounding box size due to the direct extraction of features in source domain or direct alignment of features in source and target domains, which makes the network transfer to the target domain simply by certain features in the source domain such as object contours, textures, etc., to do the object detection task. For Fig. 4(g)–(h), since these two methods may have not completely separated the DIF and DSF and the inefficient adversarial DA operation, there is also a failure to accurately detect the object bounding box size.

In another more complex inshore scene in Fig. 5, some objects with ship shape but not ship appear in SAR image, which results in some wrong detection for the feature alignment-based adversarial DA method. The imaging of the largest ship in Fig. 5 is not very clear, which prevents all methods except DTA and the proposed method from accurately detecting the object. From the experimental results, it can be seen that the proposed method has a good capability in both accurate detection of the object and correct detection of the object size.

In S → G task, we present the result of a inshore complex scene with multiship objects detection. As can be seen from ground truth Fig. 6(j), this is a harbor scene with fifteen ships spread out along the coast. Although all the methods show different degrees of wrong detection and false alarm, the proposed method still has the best detection performance. Specifically, the feature alignment-based methods result in false detection and errors in the bounding box size for

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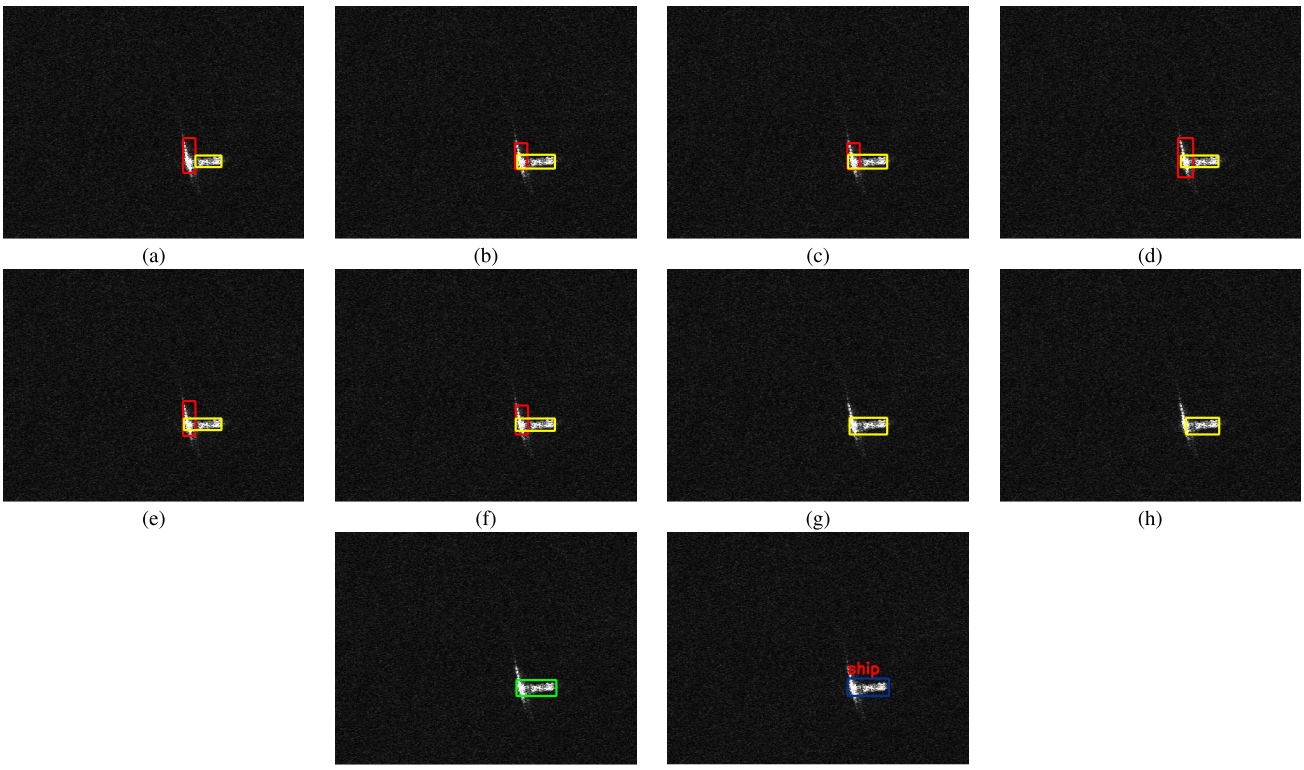


Fig. 4. Detection results of a scene with overlapping objects on the sea surface in G → S task. In (a)–(i), green indicates a correctly detected object, red indicates a wrong detected object, and yellow indicates a correctly detected object but the object's rectangular box is the wrong size. (a) Faster RCNN. (b) FACL. (c) UDA. (d) SDA. (e) SCL. (f) MeGA. (g) DTA. (h) I^3NET. (i) Proposed method. (j) Ground truth, where the object class and rectangular box is marked.

TABLE V
COMPUTATION TIME OF PER IMAGE FOR VARIOUS METHODS

Method	**Faster RCNN** [37]	**FACL** [18]	**UDA** [19]	**SDA** [44]	**SCL** [26]	**MeGA** [45]	**DTA** [30]	**I^3NET** [47]	**Ours**
G → S	209ms/img	171ms/img	174ms/img	180ms/img	182ms/img	179ms/img	173ms/img	173ms/img	170ms/img
S → G	209ms/img	172ms/img	174ms/img	178ms/img	180ms/img	179ms/img	173ms/img	173ms/img	172ms/img

small objects and objects that are close together. In contrast, the proposed method has better detection results for small objects or overlapping objects caused by sea clutter and noise. Meanwhile, the method based on extracting DIF is useful for regression task in object detection, which further proves our view about extracting DIF and DSF is helpful for SAR images object detection.

In addition, we compare the detailed computation times of all methods in both tasks. The comparison results are shown in Table V, where we compare the entire method by comparing the computation time of each image, where "img" represents each image. As can be seen from Table V, the proposed method also has a good capability in terms of computational time and does not increase the computational burden of the device due to the designed DA subnetwork.

C. Ablation Studies

To evaluate our proposed method more clearly and thereby to have a clearer perception of the effectiveness of the proposed method, it is further analyzed in this section by means of ablation experiments.

1) DIF and DSF Extraction: Compared to previous DA object detection methods that directly use domain adversarial learning strategy to perform feature alignment in source and target domains, we propose that the object features in both domains contain domain-invariant and domain-specific portions, while the latter is more suitable for performing feature alignment and the former is more suitable for performing regression localization of objects. In this ablation study, we perform three ablation experiments on the proposed network framework without DIF and DSF extraction, DIF and DSF extraction only on local features, and DIF and DSF extraction only on global features to verify the effectiveness of this operation.

TABLE VI
DETECTION RESULTS (%) OF CONVENTIONAL DECOMPOSITION METHOD AND PROPOSED DECOMPOSITION METHOD

Method	G → S			S → G		
	PR	RE	mAP	PR	RE	mAP
Without	72.6	79.7	76.3	69.5	78.6	68.0
Local features only	75.1	82.8	78.3	70.3	80.2	70.3
Global features only	76.4	83.7	79.2	72.8	80.8	72.9
Proposed	**78.4**	**86.3**	**81.5**	**73.7**	**81.9**	**74.4**

From Table VI, it is obviously seen that none of these three cases has a better detection performance than the overall method we have proposed. Meanwhile, combining Tables IV and VI, we can obtain that even the operation of

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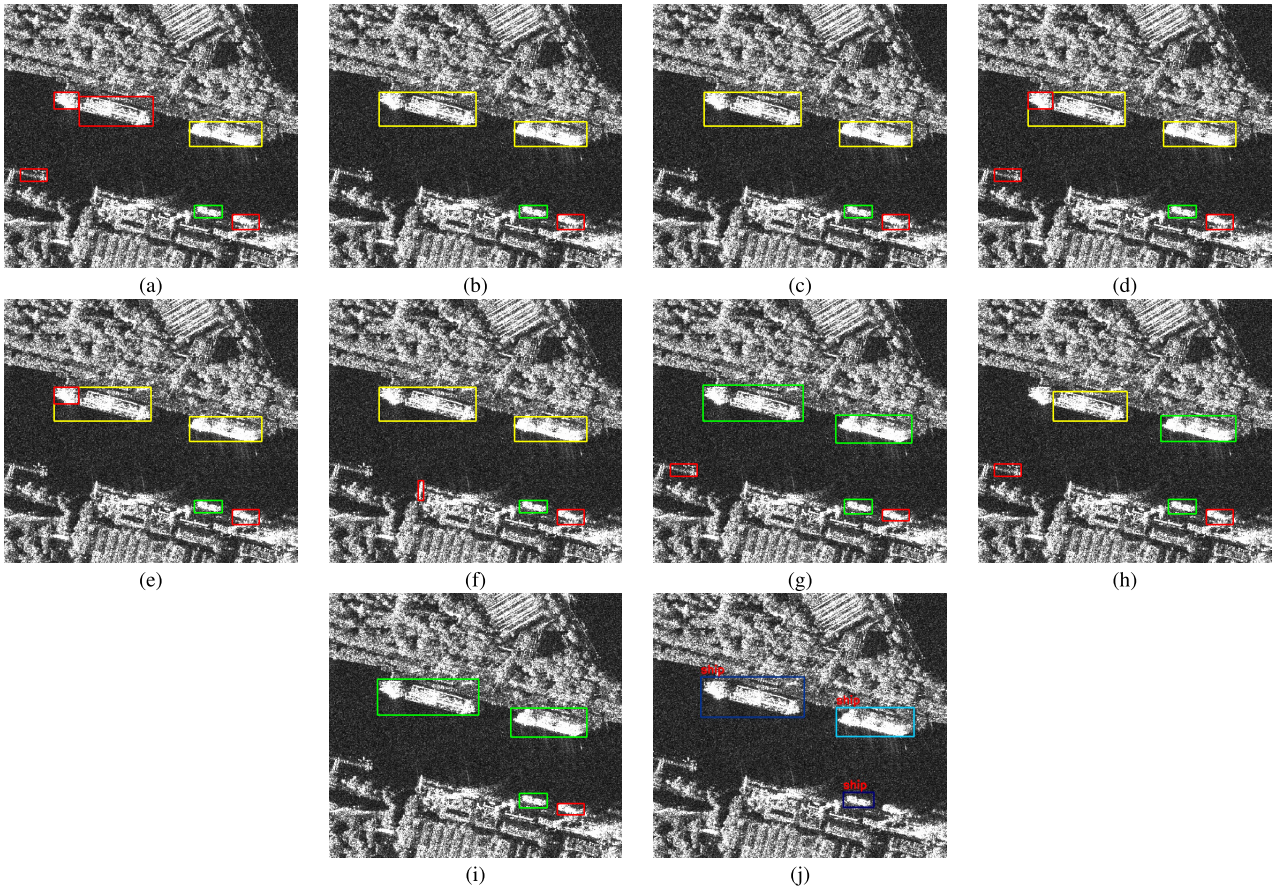


Fig. 5. Detection results of a scene with multi objects inshore in G → S task. In (a)–(i), green indicates a correctly detected object, red indicates a wrong detected object, and yellow indicates a correctly detected object but the object's rectangular box is the wrong size. (a) Faster RCNN. (b) FACL. (c) UDA. (d) SDA. (e) SCL. (f) MeGA. (g) DTA. (h) I^3NET. (i) Proposed method. (j) Ground truth, where the objects class and different colored rectangular boxes for different objects are marked.

extracting DIF and DSF only on local features is higher than the mAP of our previous feature alignment-based adversarial DA object detection method FACL [18], which rightfully proves the validity of the idea we have introduced. Fig. 7 shows the results of this ablation experiment in the form of detection results and heat maps in G → S task. For the two cases of local features only and global features only, we visualize the extracted DIF; while for the proposed method, we visualize the global feature extraction DIF. Since we do not perform DIF extraction in the first case, we do not show its heat map. It can be clearly seen in Fig. 7 that the DIF extracted by the proposed method are indeed pixels with higher heat values appearing at the contour locations of the ship objects, which means that DIF in the source and target domains are indeed features such as the contours of the ship objects. Furthermore, the upper right corner of this SAR image slice we chose is empty, so there are obvious heat values in both DIF extraction cases, but the proposed method effectively avoids this problem. Therefore, the proposed feature decomposition-based method can improve the performance of DA SAR image object detection.

2) Decomposition Method of DIF and DSF: In Section III-A, we have already discussed the decomposition of DIF and DSF by constructing DIFE and DSFE with two different convolution kernel dimensions and then making them independent by constructing objective function in conventional methods; our method is to separate DIF and DSF by vector decomposition and then construct orthogonal loss function to make them independent. In this part, we compare the detection accuracy of these two feature decomposition methods in two tasks. During this ablation experiment, we only change the decomposition of the local and global features from the network to the existing method via DIFE and DSFE [30], and the rest of the settings remain unchanged. The experimental results are shown in Table VII. In Table VII, it can be seen that the detection mAP of the proposed feature decomposition method is 2% and 3.3% higher than the conventional method in two tasks, respectively.

TABLE VII
DETECTION RESULTS (%) OF CONVENTIONAL DECOMPOSITION METHOD AND PROPOSED DECOMPOSITION METHOD

Method	G → S			S → G		
	PR	RE	mAP	PR	RE	mAP
Conventional	76.2	83.5	79.5	71.5	81.2	71.1
Proposed	**78.4**	**86.3**	**81.5**	**73.7**	**81.9**	**74.4**

3) IoU Threshold: The IoU threshold is an important aspect in object detection since it reflects the quality of detection. It is known that a higher IoU threshold indicates a bigger overlap with the ground-truth box from previous experience.

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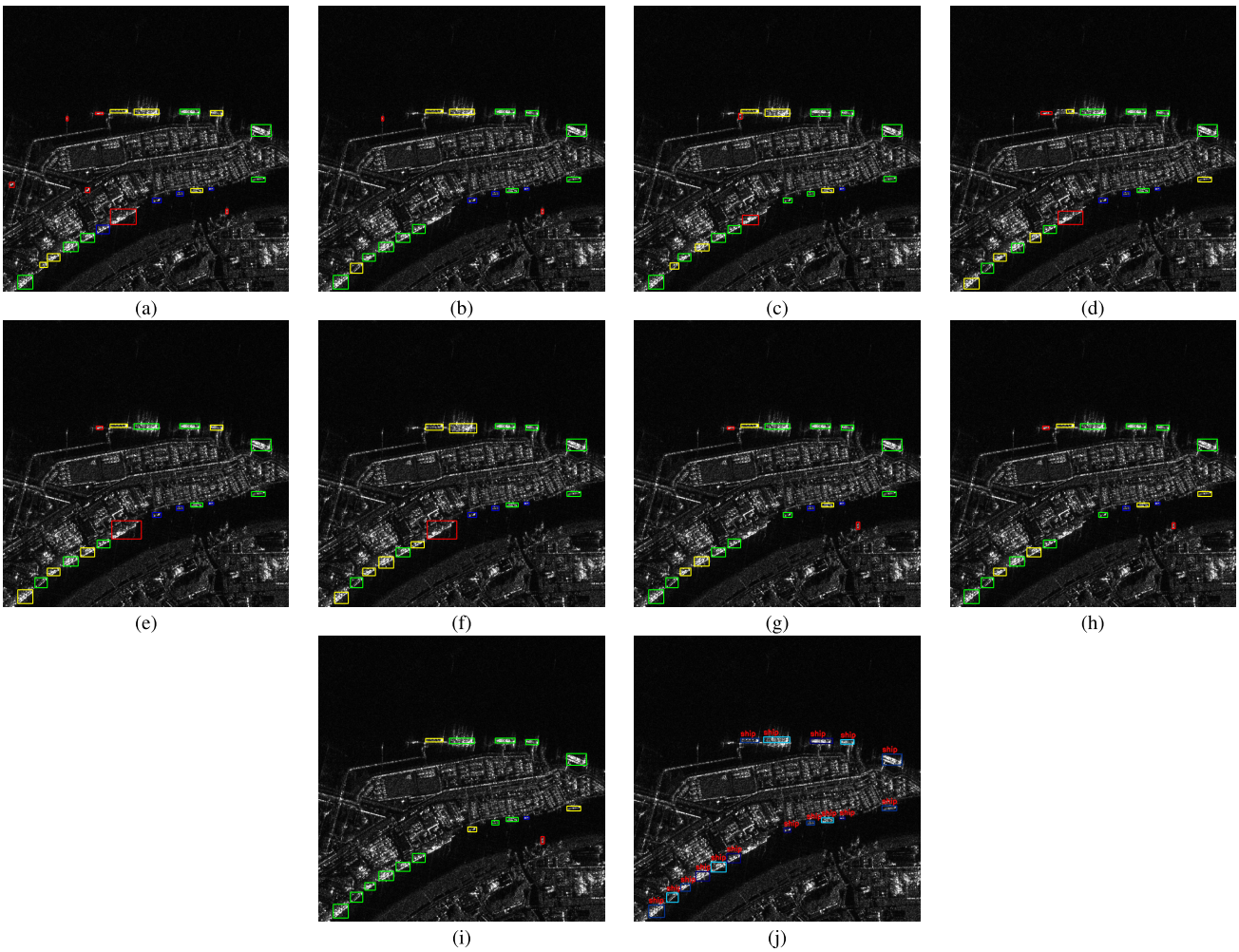


Fig. 6. Detection results of a scene with multiobjects inshore in S → G task. In (a)–(i), green indicates a correctly detected object, red indicates a wrong detected object, blue indicates a false-alarm detected object, and yellow indicates a correctly detected object but the object's rectangular box is the wrong size. (a) Faster RCNN. (b) FACL. (c) UDA. (d) SDA. (e) SCL. (f) MeGA. (g) DTA. (h) I^3NET. (i) Proposed method. (j) Ground truth, where the objects class and different colored rectangular boxes for different objects are marked.

In the above experiments, the IoU threshold is set to 0.5 in reference to almost all the previous studies. To further analyze the effect of IoU thresholds, we plot the detection performance of different IoU values on two DA object detection tasks. In Fig. 8, we can observe that the detection mAP decreases sharply with increasing IoU thresholds due to the increasingly stringent requirements for object localization quality in object detection task. Simultaneously, it should be emphasized that the proposed method consistently outperforms the corresponding other detection methods in terms of IoU thresholds. This shows that our method can conduct object detection with greater robustness and better accuracy.

4) Backbone Network: As the classification performance of backbone networks increases on the ImageNet classification task, the performance of object detectors based on these backbone networks also increases. We choose ResNet-101 as the backbone network for the initialization of the experiments, but existing studies have also employed VGG-16 as the backbone network for object detection. Here we verify the detection performance of the two backbone networks by ablation experiment whose results are shown in the Table VIII. It can be seen that the SAR image object detection mAP for the two tasks is 2.1% and 1.7% higher when the backbone network is ResNet-101 than VGG-16, respectively.

TABLE VIII
DETECTION RESULTS (%) OF RESNET-101 AND VGG-16

Method	G → S			S → G		
	PR	RE	mAP	PR	RE	mAP
ResNet-101	**78.4**	**86.3**	**81.5**	**73.7**	**81.9**	**74.4**
VGG-16	77.6	85.5	79.4	72.1	79.3	72.7

Since the object detection model with VGG backbone network has fewer parameters and a lighter model framework, it is difficult for the model with VGG backbone network to extract deeper semantic features. Meanwhile, due to the limitation of VGG's own network layers, the detection performance is relatively lower. On the contrary, since ResNet has deeper convolutional layers to make it better at extracting multilevel features of the object, which is in line with the original intention of DA methods for feature extraction. Besides, the ResNet-101 network helps to solve the problem of gradient disappearance and gradient explosion, allowing us to train deeper DA networks while ensuring good performance.

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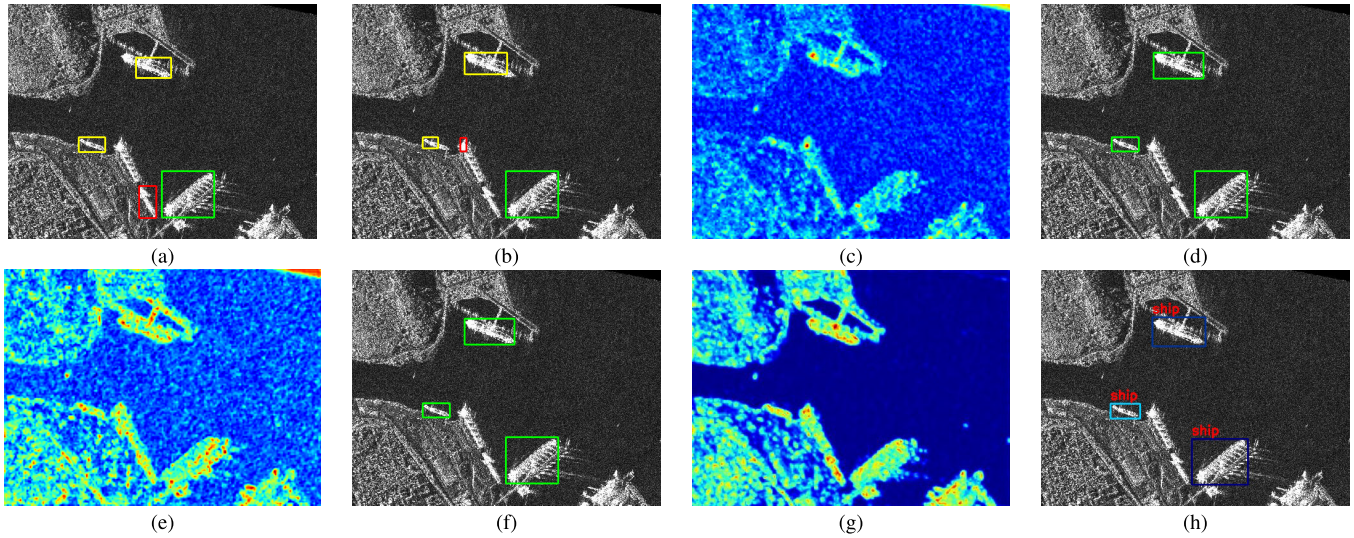


Fig. 7. Detection results and feature heat maps of a scene in G → S task. In (a)–(i), green indicates a correctly detected object, red indicates a wrong detected object, and yellow indicates a correctly detected object but the object's rectangular box is the wrong size. (a) Detection result of without DIF and DSF extraction. (b) Detection result of DIF and DSF extraction in local features only. (c) Heat map of DIF and DSF extraction in local features only. (d) Detection result of DIF and DSF extraction in global features only. (e) Heat map of DIF and DSF extraction in global features only. (f) Detection result of proposed method. (g) Heat map of proposed method. (h) Ground truth, where the objects class and different colored rectangular boxes for different objects are marked.

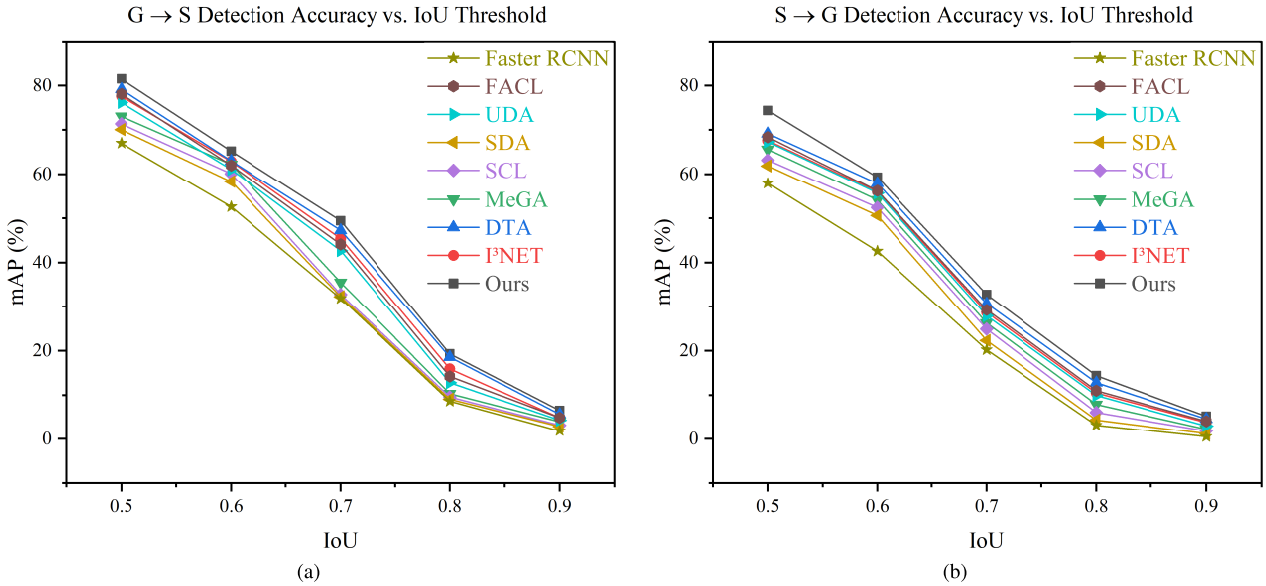


Fig. 8. Detection accuracy with different IoU thresholds. Different methods are separately indicated by different colors and symbols, where the colors and symbols correspond to the methods shown in the legend. (a) G → S task. (b) S → G task.

TABLE IX

HYPERPARAMETER SENSITIVITY ANALYSIS ON BOTH TASKS DETECTION ACCURACY (%)

γ	0.5					λ	1				
λ	0.25	0.5	0.75	1	1.25	γ	0.1	0.3	0.5	0.7	0.9
G → S **mAP**	78.9	80.2	81.0	**81.5**	79.4	G → S **mAP**	79.3	79.9	**81.5**	80.2	78.4
S → G **mAP**	72.6	73.3	73.9	**74.4**	73.7	S → G **mAP**	73.0	73.6	**74.4**	73.8	72.9

5) Hyperparameter Sensitivity: In the aforementioned experiments, we have relied on empirically chosen hyperparameters λ and γ. To verify the hyperparameter sensitivity in detail, we first analyze both hyperparameters. The parameter λ is the weight that controls the alignment of DSF features in DA method, while the parameter γ is the weight that controls DIF and DSF decomposition and their independence from each other. Table IX shows the detection results for the two tasks with the different hyperparameters. From Table IX, although we have not experimented with all combinations of these two parameters, it is still possible to conclude empirically that the best detection performance can be achieved with the hyperparameter values we set.

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V. Conclusion

In this article, we propose a feature decomposition-based method for automatic ship detection of SAR images crossing different satellites. We employ Faster RCNN as the baseline network, ResNet-101 as the backbone network, and divide the feature extraction layer of the backbone network into low level and high level. For the local features extracted from the low level and the global features extracted from the high level, DIFEs with the same structure but different parameters are proposed to extract the domain invariant portion of them, and then the local features and global features are decomposed into DIF and DSF by vector decomposition, respectively. The DSF of the local features are aligned by adversarial DA attention for instance-level features, and the DIF are then fed into the backbone network for high-level global feature extraction. The DSF in the global features are also aligned by the same adversarial DA attention, and the instance-level features from two level are fused into a new feature vector which is subjected to adversarial DA classification by GRL. Furthermore, the results of RoI alignment by RPN combined with the results of adversarial DA classification can be used to obtain accurate SAR image object detection bounding box and object class. During the experiment, Gaofen-3 and SSDD SAR image datasets obtained from several different satellites are considered as the labeled source domain and unlabeled target domain for training, respectively, and the training model is tested on the corresponding target domain. Experimental results show that the detection performance of the method is significantly better than that of the baseline network and other DA SOTA methods, which reflects the effectiveness and superiority of the method.

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